Cabarrus County's Vanishing Farmland

A decade of land cover change and what it means for the county's future

Between 2011 and 2021, Cabarrus County converted roughly 2,241 acres of farmland to other uses, shrinking its agricultural land base from 54,777 acres to 52,537 acres. Development absorbed most of that loss, adding 5,370 acres of roads, subdivisions, and commercial land over the same period. The same suburban expansion that pushed Cabarrus into North Carolina's top growth rankings (Data Brief No. 1) has steadily consumed the county's fields.

−2,241 ac

Farmland lost 2011 to 2021, from 54,777 acres to 52,537 acres

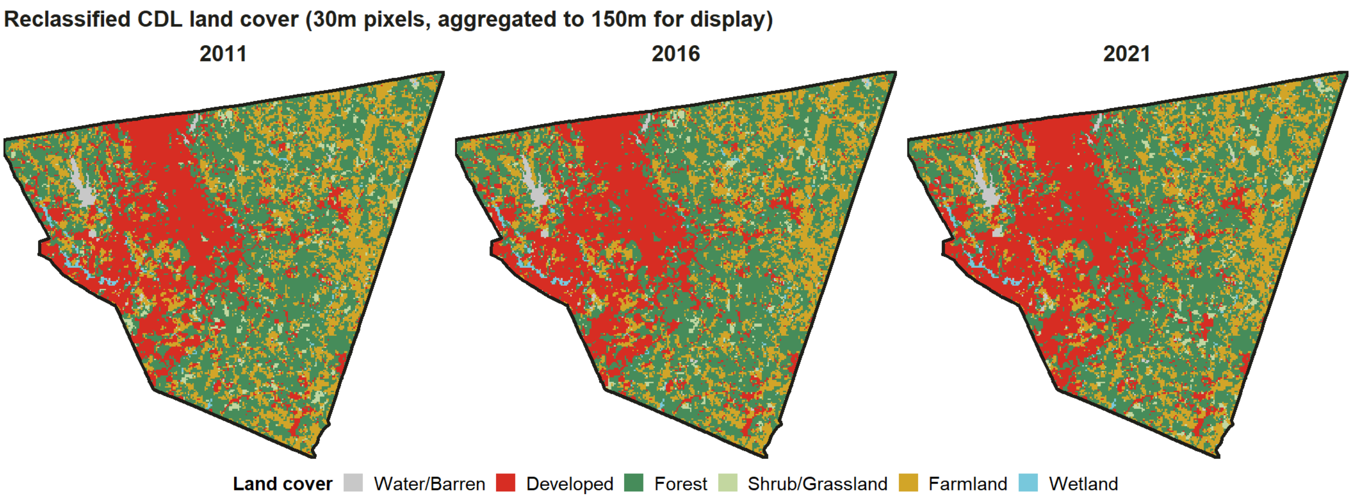
−4%

Share of 2011 farmland converted to other uses by 2021

+5,370 ac

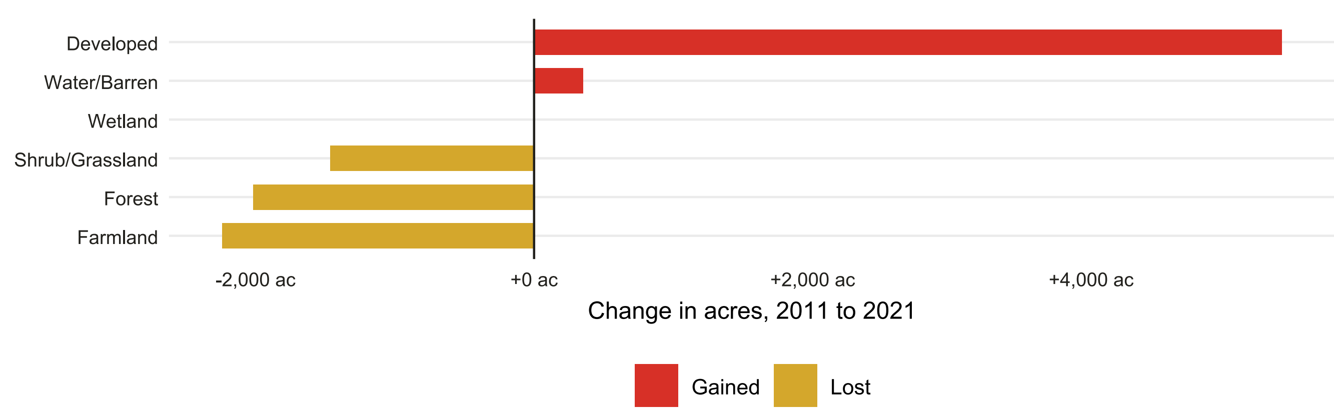
Developed land added 2011 to 2021, accounting for most of the farmland conversion

Land Cover Change Across Cabarrus County, 2011-2021



Source: USDA NASS Cropland Data Layer via CropScape. County boundary: U.S. Census Bureau TIGER/Line.

What Was Lost and What Was Gained, 2011-2021



Other/Water class excluded. Source: USDA NASS Cropland Data Layer 2011 and 2021.

! Important

The development equation. Every acre of developed land that appeared in Cabarrus County between 2011 and 2021 came from somewhere. Farmland, forest, and shrub/grassland all contributed to the conversion. This pattern mirrors the county's population trajectory documented in Data Brief No. 1: the growth that made Cabarrus a top-ranked hot spot for in-migration was built, in part, on converted open and agricultural land.

i Note

About this brief. Land cover data are from the USGS/MRLC National Land Cover Database (NLCD), a 30-meter-resolution product derived from Landsat imagery using consistent methodology across all release years. Unlike the USDA Cropland Data Layer, which is calibrated specifically for agricultural classification, NLCD uses a unified classification framework that produces reliable and comparable estimates for all land cover types across years. Years used: 2011, 2016, 2021. NLCD classes were grouped into six categories: Farmland (classes 81-82, Pasture/Hay and Cultivated Crops), Developed (21-24, Open Space through High Intensity), Forest (41-43, Deciduous/Evergreen/Mixed), Shrub/Grassland (52, 71), Wetland (90, 95), and Water/Barren (11, 12, 31). Area estimates are based on pixel counts multiplied by cell area (900 m^2 per pixel) and converted to acres. Data were retrieved programmatically via the MRLC Web Coverage Service. County boundary from U.S. Census Bureau TIGER/Line shapefiles via the `tigris` R package.